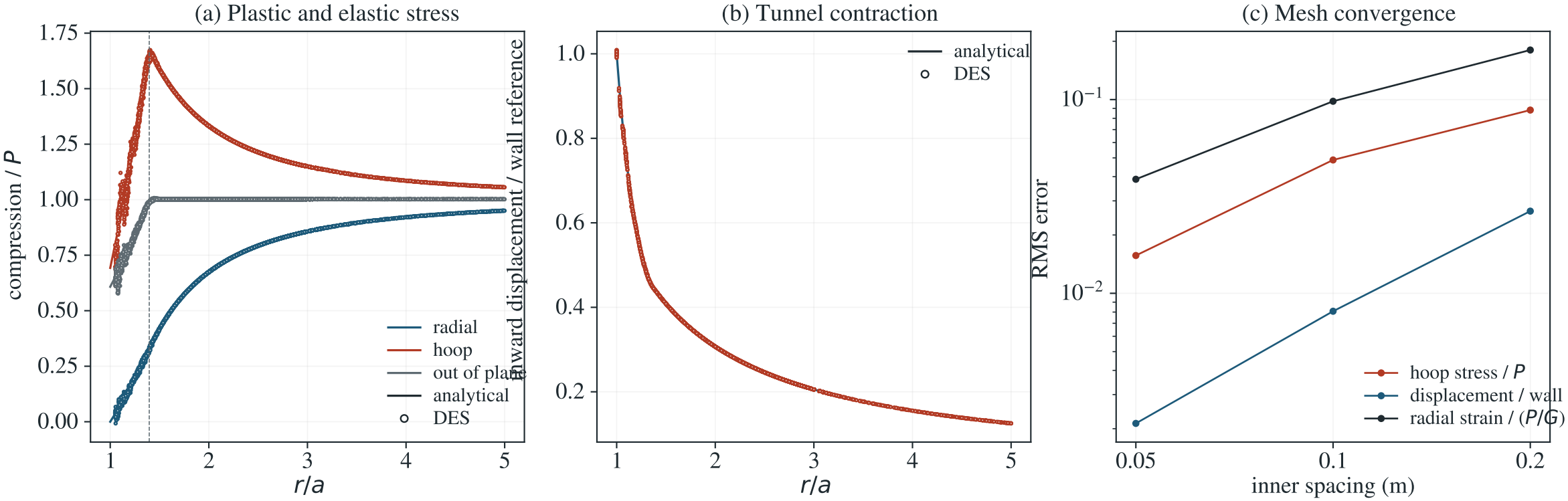

(a) Plastic and elastic stress
compression / P
inward displacement / wall reference
r/a
radial
hoop
out of plane
analytical
DES
(b) Tunnel contraction
analytical
DES
r/a
(c) Mesh convergence
RMS error
hoop stress / P
displacement / wall
radial strain / (P/G)
inner spacing (m)